

ARACHNIDS

A class within the phylum Arthropoda, arachnids include predatory spiders and scorpions, as well as mites and blood-sucking ticks.

Arachnids and related horseshoe crabs are chelicerates — arthropods named for their claw-like mouthparts (chelicerae). The head and thorax of chelicerates are fused into one body section, which carries the sensory organs, brain, and four pairs of walking legs. Unlike other arthropods, chelicerates lack antennae.

EFFECTIVE PREDATORS

Scorpions, spiders, and their relatives are land-living predators that have evolved quick ways of immobilizing and killing prey. Spiders use their

chelicerae as fangs to inject venom and many first ensnare prey by spinning webs. Scorpions inject venom with the sting in their tail. Between their legs and chelicerae, arachnids also have a pair of limb-like pedipalps. These are modified into grasping pincers in scorpions or sperm-transferring clubs in male spiders.

MICROSCOPIC DIVERSITY

Many mites are too small to see with the naked eye. They abound in almost every habitat, where they scavenge on detritus, prey on other tiny invertebrates, or live as parasites. Some lead innocuous lives in skin follicles, feathers, or fur; others cause disease or allergy. One group — the ticks — are blood-suckers and can spread disease-causing microbes.

PHYLUM	ARTHROPODA
CLASS	ARACHNIDA
ORDERS	12
FAMILIES	661
SPECIES	About 103,000



This female wasp spider waits at the centre of her dewy orb web to ensnare any insect prey that might fly in.



CHILEAN BURROWING SCORPION
Centromachetes pococki
F: Bothriuridae

A burrowing scorpion from the southern hemisphere, this species occurs in temperate forests in S. America, often under rotting logs.

4–5 cm
1½–2 in



MARITIME PSEUDOSCORPION
Neobisium maritimum
F: Neobisiidae

A common W. European coastal pseudoscorpion, this species lives under stones and seaweed in the splash and upper intertidal zones.

3 mm
⅓ in



FINGER-CLAWED PSEUDOSCORPION
Dactylochelifer latreillei
F: Cheliferidae

Pseudoscorpions are tiny arachnids that kill their prey with venomous pincers. This European species lives at the base of seashore grasses.

3 mm
⅓ in



BROAD-HEADED PSEUDOSCORPION
Chthonius ischnocheles
F: Chthoniidae

Pseudoscorpions, such as this European species, sometimes find small prey in animal nests by hitch-hiking in the fur or feathers of the parent animal.

1.5–2.5 mm
⅙–⅓ in



IMPERIAL SCORPION
Pandinus imperator
F: Scorpionidae

One of the largest of all scorpions, this African forest species is more docile and has a milder sting than many of the desert-dwellers.

15–25 cm
6–10 in



AFRICAN ROCK SCORPION
Hadogenes phyllodes
F: Hemiscorpiidae

Typical of a family of rock scorpions, this southern African species has a broad, flat body for squeezing into cracks between rocks.

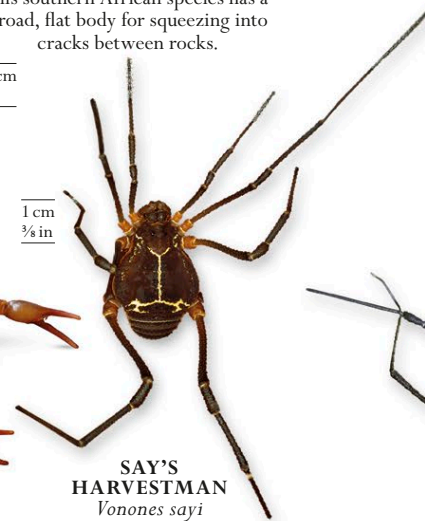
10–18 cm
4–7 in



YELLOW FAT-TAIL SCORPION
Androctonus amoreuxi
F: Buthidae

Most fat-tail scorpions are small but have potent venoms. This large species from the Sahara and Middle East has caused human fatalities.

7–10 cm
2¾–4 in



SAY'S HARVESTMAN
Vonones sayi
F: Cosmetidae

Harvestmen have an undivided body and are non-venomous. Many, including this American species, produce distasteful chemicals to deter predators.

1 cm
⅓ in



vesicle contains two poison sacs

sting

pair of eyes on top of the cephalothorax

6–8 cm
2¼–3¼ in

pedipalps modified as pincers